Lecture 35: Biological Dose, Health Physics, and Regulatory Limits

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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Introduction: The Unit Conversion Nightmare

The field of Health Physics is notorious for having two sets of units for everything (SI vs. Conventional). We must master the "Pipeline of Dose"—the sequence of converting a source term into a biological risk.

1. Source	$\xrightarrow{\text{Geometry}}$	**2. Field**	$\xrightarrow{\text{Interaction}}$	**3. Absorbed Dose**
Activity		Flux		Energy Dep.
(Ci, Bq)		(ϕ)		(Rad, Gy)

Figure 1: The physical chain of events.

1 1. The Four Tiers of Radiation Units

1.1 Tier 1: Activity (The Source)

How many atoms decay per second? This tells you nothing about danger until you know type, energy, distance and shielding.

- **Curie (Ci):** 3.7×10^{10} decays/s (Based on 1 gram of Ra-226).
- **Becquerel (Bq):** 1 decay/s (The SI unit).
- **Analogy:** The rate at which a pitcher throws baseballs.

1.2 Tier 2: Exposure (The Air)

Historically easier to measure than tissue dose. Defined by the ionization of dry air.

- **Roentgen (R):** Amount of X-ray/Gamma radiation that creates 1 esu of charge in 1 cc of dry air.
- **Analogy:** How many baseballs are flying through the air near you and how fast they are moving.

1.3 Tier 3: Absorbed Dose (The Physics)

Energy deposited per unit mass of material.

- **Rad:** 100 erg/g.
- **Gray (Gy):** 1 J/kg.
- **Conversion:** 1 Gy = 100 Rad.
- **Analogy:** How hard the baseballs hit you over time (Kinetic Energy absorbed).

1.4 Tier 4: Equivalent Dose (The Biology)

Not all energy is equal. A 1 MeV neutron does far more biological damage than a 1 MeV gamma ray.

- **Rem:** Roentgen Equivalent Man.
- **Sievert (Sv):** The SI unit.
- **Conversion:** 1 Sv = 100 Rem.

$$H(\text{Rem}) = D(\text{Rad}) \times Q$$

Where Q is the **Quality Factor** (or Radiation Weighting Factor, w_R).

2 2. Calculating Dose from Flux

How do we turn the Flux $\phi(E)$ we calculated in shielding (Lectures 33-34) into Dose rate $\dot{D}$?

2.1 The Fundamental Equation

The energy deposition rate is simply the intensity ($I = \phi E$) times the mass absorption coefficient.

$$\dot{D} = \phi \cdot E \cdot \left(\frac{\mu_{en}}{\rho}\right) \cdot C$$

Where:

- ϕ: Flux (#/cm^{2}s).
- E: Energy per particle (Joules or MeV).
- μ_{en}/ρ: Mass Energy-Absorption Coefficient (cm^2/g). This is a material property (Tissue/Water).
- C: Unit conversion constant.

2.2 Quality Factors (Q): The "LET" Concept

Why is a neutron worse than a gamma?

- **LET (Linear Energy Transfer):** dE/dx. How densely the particle deposits energy along its track.
- **Low LET (γ, β):** Sparse ionization. DNA breaks are single-strand and easily repaired by the cell. $Q = 1$.
- **High LET (α, n):** Dense ionization "furrow." Causes **Double Strand Breaks** (DSBs) in DNA. Hard to repair; leads to mutations.

Radiation Type	Quality Factor (Q)
X-Rays, Gamma, Electrons	1
Thermal Neutrons	2–5
Fast Neutrons	10–20
Alpha Particles	20

2.3 From Absorbed Dose to Biological Risk

Once the absorbed dose $\dot{D}$ is known, the equivalent dose rate is:

$$\dot{H} = \dot{D} \times Q$$

This is the final step in converting a physical radiation field into a biological risk metric.

3 3. Biological Case Studies: The "Bad Actors"

In a reactor accident, we worry about specific isotopes based on their chemistry (Bio-accumulation).

3.1 Case A: Iodine-131 (The Thyroid)

- **Chemistry:** The body cannot distinguish I-131 from stable I-127. The thyroid gland concentrates Iodine by a factor of $\approx 500\times$.
- **Hazard:** Beta emitter. Concentrates in a small gland (high dose per gram). Causes thyroid cancer (especially in children).
- **The Countermeasure (KI Pills):** Potassium Iodide.
- **Mechanism: Isotopic Dilution / Competitive Inhibition**. By flooding the blood with stable Iodine (KI) *before* exposure, you saturate the thyroid's uptake channels. Any radioactive iodine inhaled is rejected and excreted by the kidneys.

3.2 Case B: Strontium-90 (The Bones)

- **Chemistry:** Group 2 element, chemically analogous to **Calcium**.
- **Hazard:** The body builds Sr-90 into bone matrix. It has a long biological half-life.
- **Result:** Chronic irradiation of the **Bone Marrow**. Primary cause of Leukemia in high-dose scenarios.

3.3 Case C: Cesium-137 (The Whole Body)

- **Chemistry:** Group 1 element, chemically analogous to **Potassium**.
- **Hazard:** Distributes uniformly in muscle/soft tissue. Gamma emitter. Delivers "Whole Body Dose."

4 4. Acute Radiation Syndrome (ARS)

If the dose is high enough (Deterministic Effect), cells die faster than they can reproduce.

4.1 The "Dosimeter" of the Body: White Blood Cells

The first cells to vanish are the ones with the highest turnover rate: Lymphocytes.

- A drop in lymphocyte count is the standard clinical method to estimate dose immediately after an accident.

4.2 The Progression

1. **Prodromal Phase (0-48 hrs):** Nausea, vomiting. The severity/speed of onset correlates with dose.
2. **Latent Phase (The "Walking Ghost"):**
   - Patient feels better. Nausea subsides.
   - **Physiology:** The stem cells in the marrow/gut are dead, but the mature cells (RBCs, WBCs) are still circulating. The patient is "running on fumes."
   - As the mature cells die off naturally over days/weeks, they are not replaced.
3. **Manifest Illness:**
   - **Hematopoietic (>1 Gy):** Infection, bleeding (no platelets). Survivable with care.
   - **Gastrointestinal (>10 Gy):** Gut lining sloughs off. Electrolyte imbalance, sepsis. Nearly always fatal.
   - **CNS (>50 Gy):** Immediate nervous system collapse.

5 5. Regulatory Limits (10CFR20)

The Nuclear Regulatory Commission (NRC) sets limits based on the **LNT (Linear No-Threshold)** hypothesis.

5.1 The Numbers

5.2 ALARA (As Low As Reasonably Achievable)

It is not enough to stay below the 5 Rem limit for workers in the nuclear industry. If you can do a job for 100 mrem, but you get sloppy and take 200 mrem, you are in violation of ALARA, even though you are legally "safe" below 5000.

Category	**Limit (Rem/yr)**	**Notes**
Average Background	0.300–0.600	Includes Radon, Cosmic, Medical.
General Public	0.100	**ADDITIONAL** dose above background. (e.g., limit for a plant's neighbor).
Occupational (Worker)	5.000	(5000 mRem). Federal Legal Limit.
Occupational Declared Pregnant	0.500	For the gestation period (protects fetus).
LD 50/30	≈ 400	Lethal Dose to 50% in 30 days (Acute).

5.3 Dose Context: Where you Live Matters

Exposure doesn't occur just due to nuclear reactors - in fact, background radiation is generally much greater! This background radiation is not constant either. It varies wildly by location and lifestyle.

Case Study: The Background Radiation Variable

1. Sea Level vs. Altitude (Cosmic Dose):
People in **Denver, CO** (1 mile high) receive ≈ 50 mrem/yr from cosmic rays, nearly double the dose of someone at sea level (≈ 26 mrem/yr).

2. Indoor vs. Outdoor (Radon Dose):
Surprisingly, being **Indoors** usually results in a higher dose than being outdoors.

- **Outdoors:** Radon gas dilutes into the atmosphere.
- **Indoors:** Radon seeps from the ground and accumulates in basements.
- **Result:** A person staying inside a poorly ventilated basement can receive > 1000 mrem/yr, far exceeding the limit for a nuclear plant neighbor.

5.4 The Radioactivity Within: Internal Background Dose

Every human being is naturally radioactive. The food we eat, the water we drink, and the air we breathe contain primordial and cosmogenic isotopes that are continually incorporated into our tissues and bones. According to NCRP Report No. 160, the average individual receives a continuous internal dose of approximately **29 mrem/year** strictly from ingested isotopes (not including Radon, which is both variable and much larger on average). This internal dose is dominated by three sources:

- **Potassium-40 (K-40) [$\approx$ 15 to 18 mrem/yr]:** This is the absolute heavyweight of internal radiation. Potassium is an essential electrolyte required for human muscle and nerve function, and the body maintains strict homeostatic control over its concentration (roughly 140 grams in an adult). Because 0.012% of all natural potassium is the radioactive isotope K-40, your body contains a constant inventory of it. It undergoes both beta decay and electron capture, emitting a highly penetrating 1.46 MeV gamma ray that often escapes the body entirely (meaning you are constantly irradiating the people standing next to you - but of course they are irradiating you too).
- **Uranium/Thorium Series (Po-210 and Pb-210) [$\approx$ 10 mrem/yr]:** Trace amounts of Uranium and Thorium are present in all soil and water. As they decay, they produce Lead-210

and Polonium-210, which are absorbed by plants and marine life. Ingested via leafy greens, root vegetables, and seafood, these isotopes deposit in human bone and soft tissue. Because Polonium-210 is a high-energy alpha emitter, it delivers a disproportionately high biological dose despite its microscopic physical mass.

- **Carbon-14 (C-14) [≈ 1 mrem/yr]:** Created continuously in the upper atmosphere by cosmic ray interactions with nitrogen, C-14 is oxidized into carbon dioxide and incorporated into the global food web via photosynthesis. Because the body is approximately 18% carbon by mass, we contain a significant inventory of C-14. However, because it is a low-energy pure beta emitter ($E_{max} = 0.156$ MeV), the actual dose delivered to the tissue is remarkably small compared to K-40.

Engineering Perspective: The "Banana Equivalent Dose"

Because of the natural presence of Potassium-40 (K-40), foods rich in potassium are measurably radioactive. This led to the informal, educational metric known as the **Banana Equivalent Dose (BED)**.

When calculating the BED, health physicists use the **Committed Effective Dose Equivalent (CEDE)**. This calculation integrates the deposited energy over the **biological half-life** of potassium (≈ 30 days), ignoring the massive **physical half-life** of K-40 (1.25×10^9 years). This yields a calculated internal dose of roughly **0.01 mrem** (0.1 μSv) per banana. At this theoretical rate, a person would need to consume 500,000 bananas in a single year to reach the NRC's annual occupational dose limit (5,000 mrem).

However, biologically and thermodynamically, the BED is fundamentally flawed. The human body tightly regulates its potassium concentration through **homeostasis**. When you eat a banana, the fresh potassium temporarily exceeds your homeostatic limit, and your kidneys immediately excrete an equal amount to restore equilibrium. Because the new K-40 simply displaces existing K-40, your long-term radioactive burden does not increase, and the net cumulative dose from the banana is functionally zero.

5.5 Medical Dosages: From Diagnostics to Treatment

When discussing medical radiation, we span an enormous range of magnitudes—from a fraction of background radiation to doses that would be universally fatal if not strictly controlled. It is critical to distinguish between *diagnostic* procedures (low-dose, focused on minimizing stochastic risk) and *therapeutic* procedures (massive-dose, designed for deterministic cell death).

Diagnostic Doses (mrem)

Diagnostic imaging aims to use the ALARA principle (where still applicable) to achieve a clear image with the minimum possible dose. These are whole-body equivalent doses, allowing us to compare the risk directly to natural background radiation.

Therapeutic Doses (Rads)

In radiation oncology, the goal is the targeted eradication of malignant tissue. Because we are causing deterministic cell death, the concept of a "whole-body equivalent" (Rem) breaks down. Instead, oncologists prescribe the physical absorbed dose in **Rads** (or Grays, where 1 Gy = 100 Rads).

Procedure	Approx. Dose (mrem)	Notes
Extremity X-ray (Arm/Leg)	0.1	Almost negligible.
Dental X-ray	0.5	Highly localized, short duration.
Chest X-ray	10.0	Roughly equivalent to 10 days of background.
Mammogram	40.0	Soft tissue requires slightly more exposure.
CT Scan (Head)	200.0	Multiple X-ray slices compiled into 3D.
CT Scan (Abdomen)	800.0	Dense organs and larger cross-sectional volume.
PET/CT Scan	≈ 2500.0	Internal injected isotope (e.g., F-18) plus CT scan.

Table 1: Typical diagnostic effective doses. For context, the average annual natural background is ≈ 310 mrem.

A standard curative regimen for a solid tumor (such as prostate, breast, or head/neck cancer) requires a localized dose of **4,000 to 8,000 Rads** (40 to 80 Gy).

The Treatment Paradox: Surviving 8,000 Rads

Earlier, we established that the acute LD 50/30 (Lethal Dose to 50% of the population) is ≈ 400 Rads. So, how can a cancer patient survive an 8,000 Rad treatment regimen? The answer relies on two core biological and engineering principles:

1. **Strict Localization:** The 400 Rad lethal limit assumes a *whole-body* exposure, which fatally destroys the heavily dividing cells of the bone marrow and gastrointestinal lining. In modern radiotherapy, advanced Linear Accelerators (LINACs) or Proton beams use intersecting angles to confine the 8,000 Rads strictly to a tumor volume sometimes no larger than a golf ball. The rest of the patient's body receives a much smaller dose.

2. **Fractionation:** The massive dose is never delivered all at once. It is divided into daily "fractions" of roughly 200 Rads spread over 4 to 8 weeks. Healthy tissue has robust DNA double-strand repair mechanisms and can recover over the weekend. Cancer cells, however, are biologically optimized for rapid, chaotic division rather than repair; they cannot fix the radiation damage in time and undergo apoptosis (cell death).

3. **The Race:** The goal of both radiation therapy and chemotherapy is to kill the cancer cells faster than the nearby healthy cells either relying on the repair mechanism differential or (in the case of chemo) specific chemical targeting. The better your targeting (via beam design, energy levels, or chemical interactions) the more likely the treatment will be successful.

6 Case Study: The "Perfect" Poison (The Litvinenko Assassination)

In November 2006, Alexander Litvinenko, a former Russian security officer, was assassinated in London using **Polonium-210**. This event serves as a grim but definitive case study in the engineering distinction between **External Hazard** and **Internal Dose**.

6.1 The Physics of the Weapon: Polonium-210

Polonium-210 (^{210}Po) is a nearly pure alpha emitter produced by neutron bombardment of Bismuth-209 in a high-flux nuclear reactor.

- **Decay Mode:** Alpha decay to stable Lead-206 (^{206}Pb).
- **Alpha Energy:** 5.3 MeV.
- **Gamma Emission:** Negligible ($< 0.001\%$ of decays).
- **Half-Life:** 138 days (High specific activity, meaning a tiny mass yields a massive dose).
- **Quantities:** A lethal dose corresponds to only a few micrograms of Po-210.

6.2 The Engineering Failure: Why Detection Was Missed

Litvinenko presented with symptoms of severe radiation sickness (hair loss, immune failure), yet he was not diagnosed for weeks. Why?

6.2.1 1. The Shielding Paradox (External Detection)

Standard security measures failed because of the alpha particle's short range.

- **The Container:** The poison was likely transported in a simple glass vial or commercial beverage container.
- **The Physics:** A 5.3 MeV alpha particle travels less than 40 μm in glass. The container walls ($> 1000\,\mu$m) absorbed 100% of the radiation.
- **The Survey:** A standard Geiger-Müller counter held inches from the assassin's bag would read **Background levels**. Without gamma rays to penetrate the container, the source was invisible.

6.2.2 2. The Biological Catastrophe (Internal LET)

While harmless outside the body (stopped by the dead layer of the epidermis), alpha particles are the most destructive type of radiation once ingested.

- **High LET (Linear Energy Transfer):** Alpha particles are heavy (^{4}He nuclei) and doubly charged (+2). They plow through tissue like a bulldozer, depositing all their energy in a very short distance ($\sim 50\,\mu$m, or about 5 cell diameters).
- **The Quality Factor (Q):** In calculating dose equivalent (Sieverts), alpha particles are assigned a Weighting Factor (w_R) of **20**.

$$\text{Dose (Sv)} = \text{Absorbed Dose (Gy)} \times 20 \tag{1}$$

  This means 1 Joule of alpha energy does **20 times the biological damage** of 1 Joule of gamma or beta energy. The massive double-strand DNA breaks are almost impossible for cells to repair.

6.3 The Forensic Trace

The assassination was ultimately uncovered not by a Geiger counter, but by **Alpha Spectroscopy** on biological samples (urine). Once identified, investigators could trace the path of the poison (teapots, hotel rooms, airplane seats) because minute traces of Po-210 had escaped the container or been excreted by the assassins.

> **Engineering Lesson:** "Non-detectable" does not mean "Safe." A survey meter only tells you what is penetrating the probe wall. If you suspect an alpha hazard, you must use an open-window probe (Pancake) or take swipe samples for laboratory analysis.

References

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